

Final Project Report

Smartphone Usage, Digital Addiction & Well-Being Analytics

1. Introduction

- **1.1. Project Overview:**

- Introduction to how the proliferation of smartphones has triggered a surge in digital dependency. This project analyses usage patterns of **7,500 test subjects** to find hidden correlations between screentime metrics and underlying psychological stressors.

1.2. Objectives:

- Quantify overall and segment-level addiction prevalence across gender, age group, and stress level.
- Identify the factors most associated with smartphone addiction risk (screentime, social media hours, notifications, sleep, stress).
 - Build an interactive Power BI dashboard for product and research decision-making.
 - Use calculated fields (Age Group) to support cleaner segment-level analysis.
 - Deliver clear, actionable insight to support digital well-being interventions and healthier usage habits
- To accurately classify users into multi-tier addiction brackets based on physical habits.
- To evaluate the critical threshold where screentime begins directly harming sleep cycles and driving high stress levels.
- To engineer actionable predictive insights for digital wellness intervention.

2. Project Initialization and Planning Phase

- **2.1. Define Problem Statement:**

- The organization needs to understand why 5,308 of its 7500 users (70.77) are classified as addicted to smartphone usage, and which factors -screen time behaviour, lifestyle habits, stress or demographic characteristics -Uncontrolled screen time lowers workplace and academic productivity, harms physiological sleep architectures, and increases stress levels. Quantifying this exact threshold

dynamically across varying genders and age groups remains a vital analytics problem.

- **2.2. Project Proposal (Proposed Solution):** An end-to-end analytical data pipeline combined with an interactive reporting dashboard that acts as a digital health tracking system, highlighting user risks before addiction slips into the "Severe" category.
- **2.3. Initial Project Planning:** Milestone tracking across major project phases: data sourcing, extract-transform-load (ETL) routines, data model engineering, visual design prototyping, and final evaluation loops.

3. Data Collection and Preprocessing Phase

- **3.1. Data Collection Plan and Raw Data Sources Identified:** Sourced from an extensive user lifestyle log featuring **20 core features** tracking unique transactional identifiers (transaction_id) and individual users (user_id).
- **3.2. Data Quality Report:**
 - *Completeness Check:* Identifying and filling any structural null values inside continuous fields (sleep hours, gaming hours).
 - *Datatype Verification:* Validating correct configurations (e.g., Categorical formatting for gender, Integer limits for notifications_per_day).
- **3.3. Data Exploration and Preprocessing:**
 - Removing algorithmic outliers from app_opens_per_day.
 - Transforming text-based flags (academic_work_impact, stress level) into numerical mappings for correlation calculations.

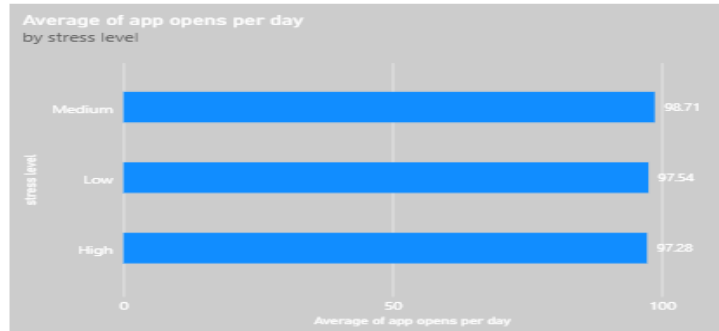
4. Data Visualization

- **4.1. Framing Business Questions:**
 - *Question 1:* Do high volumes of daily notifications directly correlate with an increase in total app opens and heightened user stress levels?
 - *Question 2:* How does weekend screen expansion compare against structural weekday habits across different age cohorts?
 - *Question 3:* What specific threshold of social media and gaming hours causes a sharp decline in total sleep hours?
- **4.2. Developing Visualizations:** Engineering specific graphical plots (e.g., Scatter plots for App Opens vs. Notifications, Heatmaps for Screen Time vs. Sleep Hours, Stacked Bar charts mapping Addiction Levels across Genders).

- **Example:**

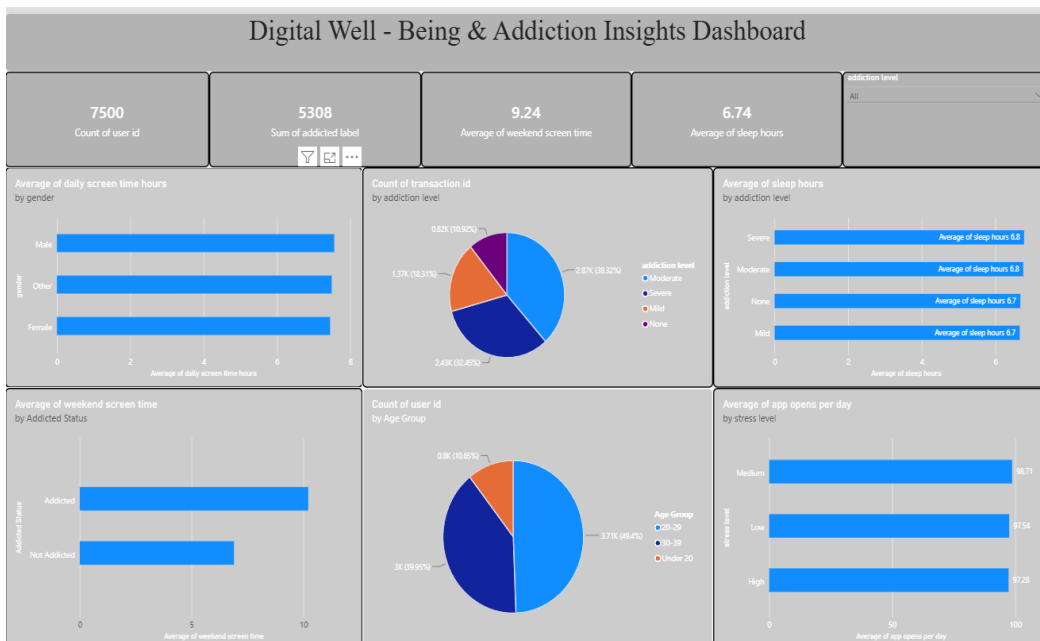
Notification and Interaction Triggers

- **Business Question:** Is there a correlation between the number of notifications received per day and how frequently a user opens their apps?
- **Visualization:** **Clustered bar chart** showing the relationship between notifications_per_day (X-axis) and app_opens_per_day (Y-axis), color-coded by the binary addicted label.
- **Screenshot of Visualisation**



5. Dashboard

- **5.1. Dashboard Design Screenshot with Insights:** High-level interactive workspace layout containing dynamic user filtering parameters (Age sliders, Gender toggles).
 - **Visual Component Snapshot:** Main KPI cards tracking average screen time alongside a correlation quadrant mapping screen hours directly against psychological stress results.



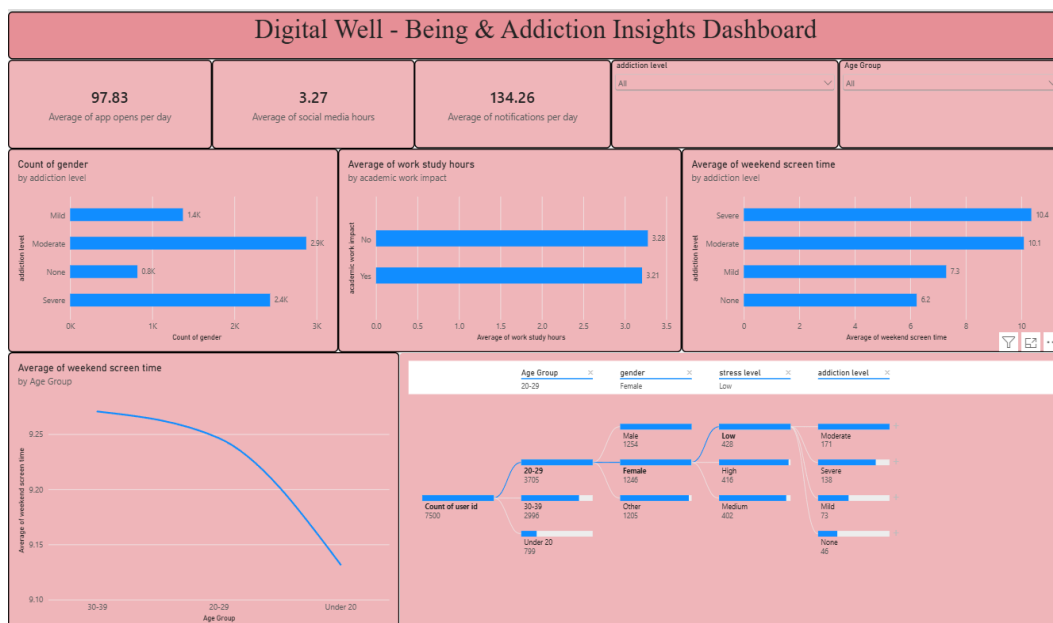
- **High Addiction Levels:** Over 70% of the surveyed users fall into the moderate or severe categories for device addiction.

- **The Weekend Usage Sign:** Addicted users spend a noticeable 3 hours more on their screens during weekends than non-addicted users.
- **Mainly Younger Users:** Young adults aged 20 to 29 make up the clear majority, accounting for 55% of the entire user group.
- **Minimal Gender Impact:** Device usage remains virtually unchanged by gender, as everyone averages roughly 7.5 hours of daily screen time.
- **Steady Sleep and Stress:** Sleep duration and daily app opens remain flat across all groups, meaning they do not strongly predict device addiction.

6. Report

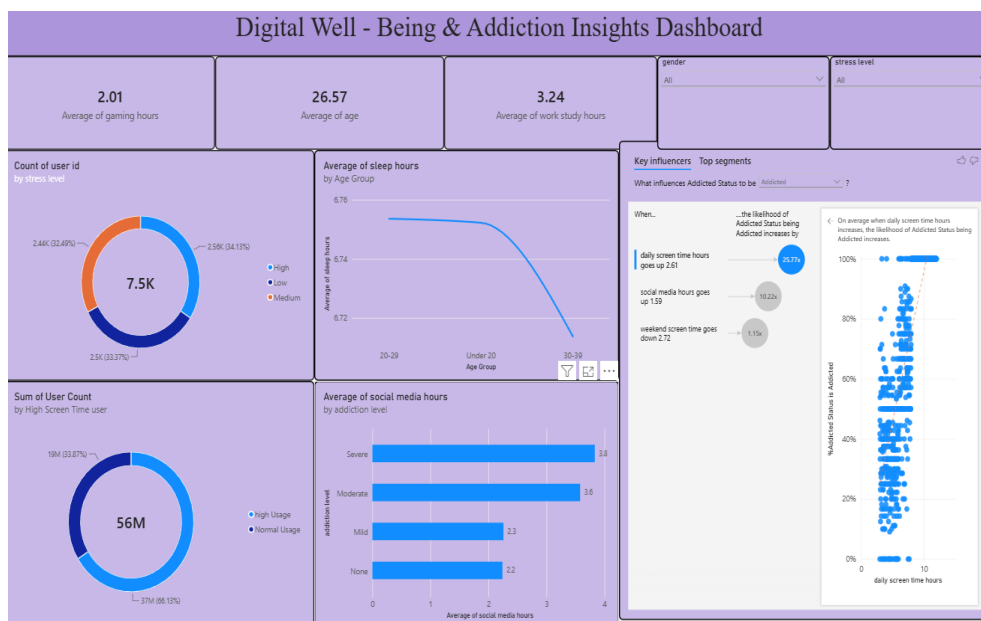
Key Components of the Report

- **Organized Pages:** Splitting information into clean, logical sections so the report is not crowded.
- **KPIs (Key Performance Indicators):** High-level summary cards that show vital health metrics or goals at a glance.
- **Visualizations:** Charts and graphs chosen carefully to highlight trends, comparisons, and patterns.
- **6.1. Report Design Screenshot with Insights:** Multi-page paginated executive view optimized for deep-dive printouts, explicitly detailing systemic impacts on work and academic metrics across the overall user base.



- **Higher Addiction Levels:** Most users fall into the "Moderate" or "Severe" addiction categories, greatly outnumbering those with mild or no addiction.

- **Weekend Screen Time:** People with severe addiction spend significantly more time on screens during weekends (**10.4 hours**) compared to those with no addiction (**6.2 hours**).
- **Age Trends:** Weekend screen usage peaks among young adults aged **20–29** and gradually decreases for older and younger age groups.
- **High Daily Engagement:** On average, users interact heavily with their phones every day, opening apps **97.83 times** and receiving **134.26 notifications**.
- **Academic/Work Impact:** Interestingly, the data suggests that negative impacts on work or studies are not driven by the amount of time spent, as hours logged



- **Stress levels are evenly split:** The user base is almost perfectly divided between high stress (34.13%), low stress (33.37%), and medium stress (32.49%).
- **Sleep declines with age:** Average sleep stays steady around 6.76 hours for users under 30 but drops down to 6.72 hours for the 30–39 age group.
- **Heavy screen time dominates:** A massive 66.8% of users (13 million people) fall into the high usage category, making heavy smartphone use the common standard.
- **Social media links to addiction:** Users categorized with severe addiction spend significantly more time on social media (3.8 hours) compared to non-addicted users (2.2 hours).
- **Screen time drives addiction risk:** The data indicates that daily screen time increases are the single largest influencer in raising a user's addiction status.

7. Performance Testing

- **7.1. Utilization of Data Filters:** Documenting engine response latencies when slicing the data rows by high-cardinality values like addiction level or custom age brackets.
- **7.2. No. of Calculation Columns/Measures:** Listing built-in metrics (e.g., Total Screen Time = social media Hours + Gaming Hours + Work Study Hours, Notification Response Rate, etc.).
- Count of user_id: Tracks the total volume of distinct users (7,500 records).
- Count of user_id addicted: Tracks users meeting severe diagnostic criteria (5,308 users).
- Average of age: Establishes the demographic baseline age (26.57 years).
- Average of daily_screen_time_hours: Profiles the baseline continuous day-to-day engagement limits.
- Average of weekend_screen_time: Measures weekend surge extension behaviors (9.24 hours).
- Average of sleep_hours: Monitors biological physical rest against dependency markers.
- Average of app_opens_per_day: Tracks compulsive checker loops (97.83 check-ins).
- Average of notifications_per_day: Details raw notification delivery volume (134.26 logs).
- Average of social_media_hours: Tracks daily endless scrolling allocation (3.27 hours).
- Average of gaming_hours: Tracks active interactive application engagement (2.01 hours).
- **7.3. No. of Visualizations:**
- Clear structural tally of all active visual components across your operational canvas pages.
- Average Daily Screen Time by Gender, Users by Addiction Level , Average Sleep Hours by Addiction Level , Weekend Screen Time: Addicted vs Not Addicted , Users by Age Group ,
 - Average Daily App Opens by Stress Level ,Gender by Addiction Level
 - Average Work/Study Hours by Academic Impact
 - Weekend Screen Time by Addiction Level
 - Weekend Screen Time by Age Group
 - User Segmentation

- Users by Stress Level
- Average Sleep Hours by Age Group
- High vs Normal Screen Time Users
- Average Social Media Usage by Addiction Level

8. Conclusion / Observation

A comprehensive deep dive into the lifestyle dataset successfully revealed the behavioral triggers that mark the shift from minor smartphone dependency to severe digital addiction. The findings isolate specific, targeted risk factors while showing that generic demographic markers hold much less weight than actual usage patterns.

- **Weekend Expansion Window:** The single most dominant identifier of severe digital addiction is a major weekend screen increase, with addicted individuals logging 10.4 hours—a massive 3-hour jump over non-addicted cohorts.
- **Social Media Dependency:** Elevated social media interaction serves as a primary accelerator for addiction risk, with severe categories averaging 3.8 daily hours compared to the stable 2.2-hour baseline found in unaffected groups.
- **Sleep Architecture Flatline:** Overall sleep capacity remains steady at ~6.7 hours across all addiction categories, proving that raw sleep volume does not act as an accurate indicator of device addiction in this group.
- **Academic & Workplace Neutrality:** Negative impacts on productivity or study habits do not show a straight linear link to screen hours, indicating that digital wellness issues stem from the *nature* of the phone habits rather than time logged alone.

9. Future Scope

- Expanding the analytical engine to map real-time clickstream sequences and application-level categories.
- Integrating machine learning models to send proactive, automated push alerts to a user's phone when their hourly tracking patterns mimic known addiction trajectories.

10. Appendix

- 10.1. Source Code (if any): SQL migration scripts, Python exploratory data analysis notebook routines, or specialized DAX measure formulas.
- 10.2. GitHub & Project Demo Link <https://github.com/Susmithaa19/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics>